Electron-Valley Resonance in monolayer Transition Metal Dichalcogenides

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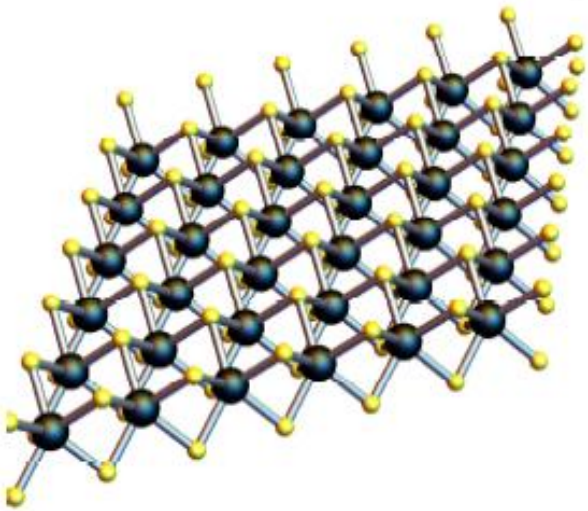
October 10, 2017

Overview

1. monolayer TMDCs
   - Lattice and bands structure
   - Tight-binding model and effective Hamiltonian
2. QD in TMDCs
   - TMDCs under magnetic field
   - Quantum dot model
   - Quantum dot model
3. Disorder and spin-valley resonance
   - Disorder and spin-valley Zeeman interaction
   - Magnetic impurity and spin-valley Zeeman interaction
   - Spin-valley resonance

Lattice structure

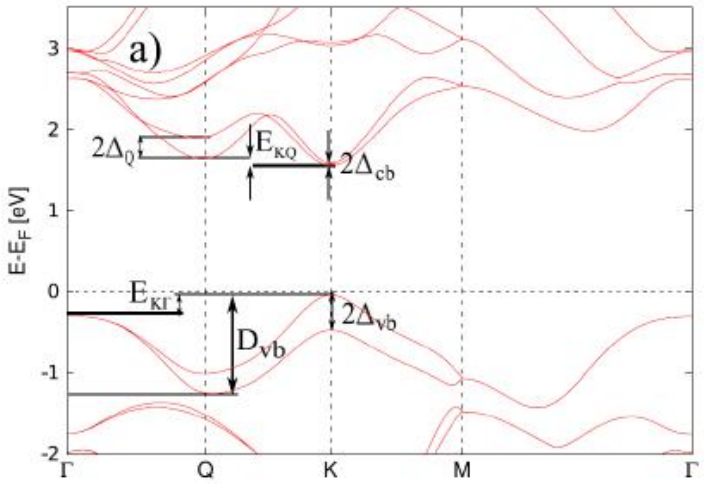
- Primitive unit-cell contains a MX_2, M is the transition metal atom of VIB family in periodic table (Mo, W,etc.) and X is a chalcogen atom (S, Se, Te).
- X-M-X sandwich structure with a layer sublattice from M and two layer sublattice from X; The honeycomb lattice from z-direction projection view.



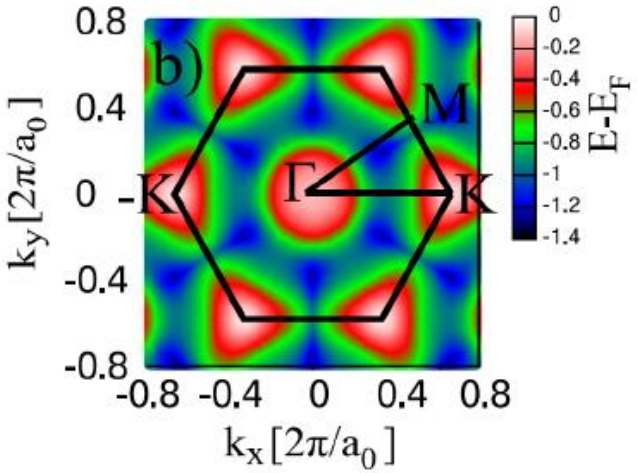


Bands structure

- hexagonal Brillouin zone (BZ)
- The maximum of valence band and minimum of conduction band are located at two-fold degenerate K points



(a) Band structure along Γ-Q-K-M-Γ



(b) Band structure in 2D BZ

Andor Kormányos *et al.* 2D Mat. (2015)

Tight-binding model

Local atomic orbitals $p^{\beta}_{\pm 1,i,s}$ ($\beta = \mathrm{S,A}$) , $d_{\pm 1,i,s}$.

On-site atomic energy

H_{atom} , on-site energy parameters $\varepsilon^{l}_{p,d}$ ($l = 1, 2$ is the angular momentum quantum number).

Hopping energy

$H_{\mathrm{hop}} = H_{dd} + H_{dp} + H_{pd}$, hopping energy parameters $t^{ij}_{\mu\nu}$ ($\mu, \nu = p, d$).

SOC

$$H_{\mathrm{so}} = \frac{\hbar}{4m_e^2c^2}\frac{1}{r}\frac{dV(r)}{dr}\mathbf{L}\cdot\mathbf{S} \ , \ \lambda_{\mathrm{M}} = \frac{\hbar}{4m_e^2c^2}\int d\mathbf{r} R^{*}_{\mathrm{M},l}(r)\frac{1}{r}\frac{dV(r)}{dr}R_{\mathrm{M},l}(r) \ ,$$
$$\lambda_{\mathrm{X}} = \frac{\hbar}{4m_e^2c^2}\int d\mathbf{r} R^{*}_{\mathrm{X},l}(r)\frac{1}{r}\frac{dV(r)}{dr}R_{\mathrm{X},l}(r)$$

Alexander J. Pearce *et al.* PRB (2016)

Bands model

In **k**-space:

$$p^{\beta}_{\mu,i,s} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}_i} p^{\beta}_{\mu,\mathbf{k},s} \ , \ d_{\nu,i,s} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}_i} d_{\nu,\mathbf{k},s}.$$

11-band model classified by σ_h symmetry $\psi_{\mathbf{k}} = (\psi^{\mathrm{E}}_{\mathbf{k}}, \psi^{\mathrm{O}}_{\mathbf{k}})^T$ where $\psi^{\mathrm{E}}_{\mathbf{k}} = (d_{z^2,\mathbf{k},s}, d_{\tau 2,\mathbf{k},s}, d_{-\tau 2,\mathbf{k},s}, p^{\mathrm{S}}_{\tau 1,\mathbf{k},s}, p^{\mathrm{S}}_{-\tau 1,\mathbf{k},s}, p^{\mathrm{A}}_{z,\mathbf{k},s})^T$ and $\psi^{\mathrm{O}}_{\mathbf{k}} = (d_{\tau 1,\mathbf{k},s}, d_{-\tau 1,\mathbf{k},s}, p^{\mathrm{A}}_{\tau 1,\mathbf{k},s}, p^{\mathrm{A}}_{-\tau 1,\mathbf{k},s}, p^{\mathrm{S}}_{z,\mathbf{k},s})$;

$$H_{\mathbf{k}} = \begin{pmatrix} H^{\mathrm{E}} & H^{\mathrm{EO}} \\ H^{\mathrm{OE}} & H^{\mathrm{O}} \end{pmatrix}, H^{\mathrm{E}} = \begin{pmatrix} H^{\mathrm{E}}_{\mathbf{k},dd} & H^{\mathrm{E}}_{\mathbf{k},dp} \\ H^{\mathrm{E}}_{\mathbf{k},pd} & H^{\mathrm{E}}_{\mathbf{k},pp} \end{pmatrix}, H^{\mathrm{O}} = \begin{pmatrix} H^{\mathrm{O}}_{\mathbf{k},dd} & H^{\mathrm{O}}_{\mathbf{k},dp} \\ H^{\mathrm{O}}_{\mathbf{k},pd} & H^{\mathrm{O}}_{\mathbf{k},pp} \end{pmatrix}.$$

The off-diagonal block matrices above originate from the spin-orbital coupling, which mix up the even and odd sectors.

Effective Hamiltonian

Energy bands near K point, known from the DFT results shown, the lowest conduction- and highest valence bands all possess an even symmetry. Therefore we focus on the even sector bands around **K** points. By expanding the dispersion around $\mathbf{K} = \left(\frac{4\pi a}{3\sqrt{3}}, 0\right)$ ($\mathbf{K}' = -\mathbf{K}$) regrading to small deviation momentum at each valley point $\mathbf{k} = \pm\mathbf{K} + \mathbf{q}$.

- Two-band model:

$$H^E_{\mathbf{q}} = \begin{pmatrix} H_{\mathrm{L}} & V^\dagger \\ V & H_{\mathrm{H}} \end{pmatrix} \xrightarrow{\text{Schrieffer-Wolff}} H^\tau_{\mathrm{eff}} = H^\tau_0 + H^\tau_{\mathrm{trig}} + H^\tau_{\mathrm{cub}} + H_{\mathrm{so}}$$

$$H_0 = \begin{pmatrix} \varepsilon_{\mathrm{vb}} & \gamma q^+_\tau \\ \gamma q^-_\tau & \varepsilon_{\mathrm{cb}} \end{pmatrix} + \begin{pmatrix} \alpha|q_\tau|^2 & 0 \\ 0 & \beta|q_\tau|^2 \end{pmatrix},$$

$$H_{\mathrm{so}} = \begin{pmatrix} \Delta_{\mathrm{vb}}\tau s_z & 0 \\ 0 & \Delta_{\mathrm{cb}}\tau s_z \end{pmatrix}.$$

Effective Hamiltonian

- One-band model: Schrieffer-Wolff transformation applied further on the two-band model. By projecting out the valence band model, one get the one-band effective Hamiltonian for conduction band

$$\Delta H_H = \frac{1}{2}\sum_{\{L\}}\left[\frac{V_{HL}V^{\dagger}_{L'H}}{E_H - E_L} + \frac{V_{HL}V^{\dagger}_{L'H}}{E_H - E_{L'}}\right]$$

$$H^{\tau,s}_{\mathrm{cb}} = H_H + \Delta H_H \approx \beta|q_\tau|^2 + \frac{\gamma^2 q^-_\tau q^+_\tau}{\varepsilon_{\mathrm{cb}} - \varepsilon_{\mathrm{vb}}} + \Delta_{\mathrm{cb}}\tau s_z.$$

TMDCs under magnetic field

The mechanical wave vector $\boldsymbol{\pi} = \mathbf{q} + \frac{e}{\hbar}\mathbf{A}$, commutator $[\pi_\tau^-, \pi_\tau^+] = \frac{2e}{\hbar}\tau B_z$ ($\pi_\tau^\pm = \tau\pi_x \pm i\pi_y$). The effective conduction band Hamiltonian:

$$H_{\text{cb}}^{\tau,s} = \frac{\hbar^2\pi_\tau^+\pi_\tau^-}{2m_{\text{eff}}} + \Delta_{\text{cb}}\tau s_z + \frac{1}{2}g_e\mu_B s_z B_z + \frac{1}{2}g_v\mu_B\tau B_z.$$

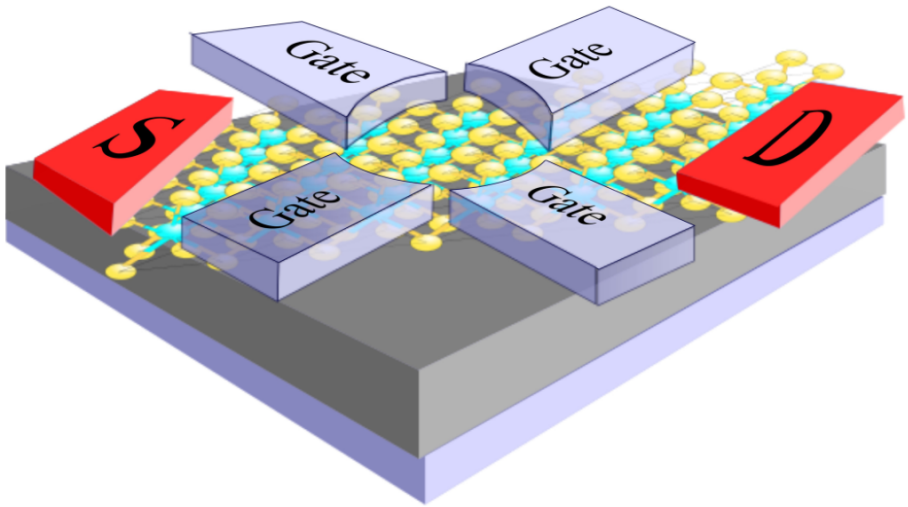
Under the symmetric gauge $A_x = -\frac{1}{2}B_z y$, $A_y = \frac{1}{2}B_z x$,
$\pi_\tau^\pm = \frac{\pm i}{l_B}\sqrt{\frac{\rho}{2}}e^{\pm i\tau\varphi}(1 \mp 2\tau\partial_\rho - \frac{i}{\rho}\partial_\varphi) = \frac{\sqrt{2}}{l_B}a_\tau^\pm$ where
$\rho = \frac{1}{2}(\frac{r}{l_B})^2 \approx \frac{B_z}{2}(\frac{r}{25\text{nm}})^2$. The Hamiltonian is then written as

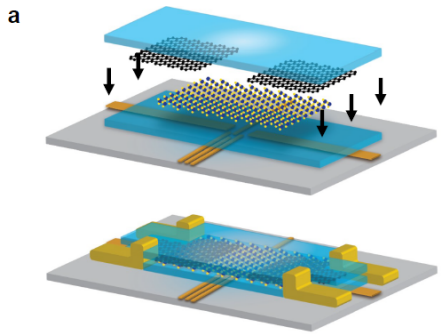
$$H = \hbar\omega_c(a_\tau^\dagger a_\tau + \frac{1}{2}) + \Delta_{\text{cb}}\tau s_z + \frac{1}{2}g_e\mu_B s_z B_z + \frac{1}{2}g_v\mu_B\tau B_z \ , \ \omega_c = \frac{eB_z}{m_{\text{eff}}}.$$

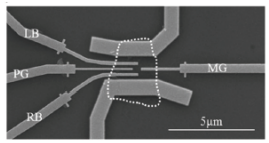
Landau-level structure Andor Kormányos *et al.* NJP (2015)

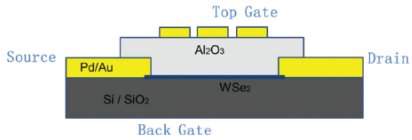
Quantum dot model

- Circular hard-wall barrier model $U_{\text{dot}}(r) = U_0\Theta(r - R_d) \quad (U_0 \to \infty)$.
- Schrödinger equation:
  $\left[\hbar\omega_c(a_\tau^\dagger a_\tau + \frac{1}{2}) + H^{\tau,s} + U(\rho)\right]\Psi^{\tau,s}(\rho,\varphi) = E\Psi^{\tau,s}(\rho,\varphi)$ with attempt wave function
  $\Psi_m^{\tau,s}(\rho,\varphi) = \frac{1}{\sqrt{2\pi}}e^{im\varphi}\Phi_m^{\tau,s}(\rho)\chi_s\ ,\ \Phi_m^{\tau,s}(\rho) = \rho^{|m|/2}e^{-\rho/2}u_m^{\tau,s}(\rho)$







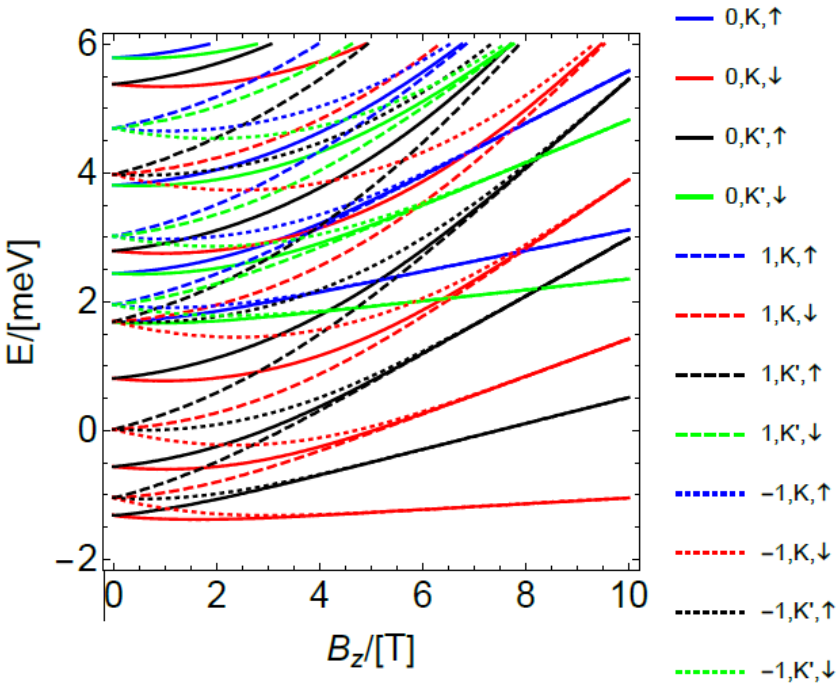


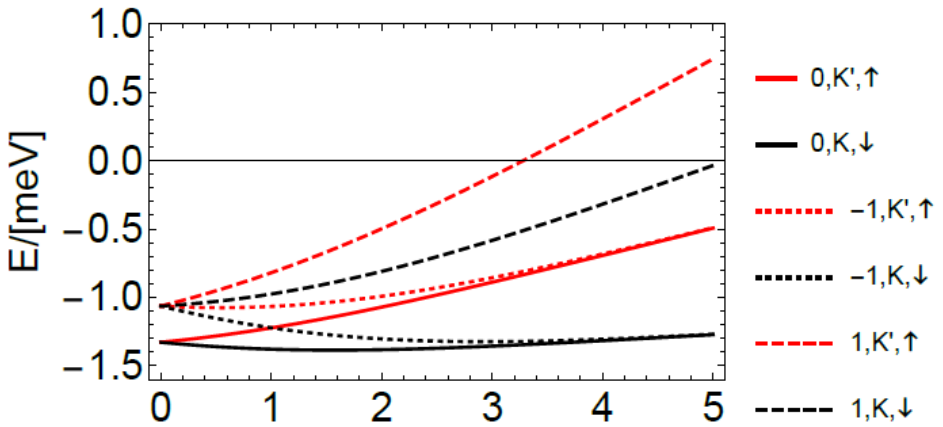
Andor Kormányos *et al.* PRX (2014);

K. Wang *et al.* arXiv:1708.08961 (2016); X. X. Song *et al.* Nanoscale (2015)

Quantum dot model

- "Kramer's pairs" are lower energy states
- At large B_z regime, QD levels closely merge into Landau levels

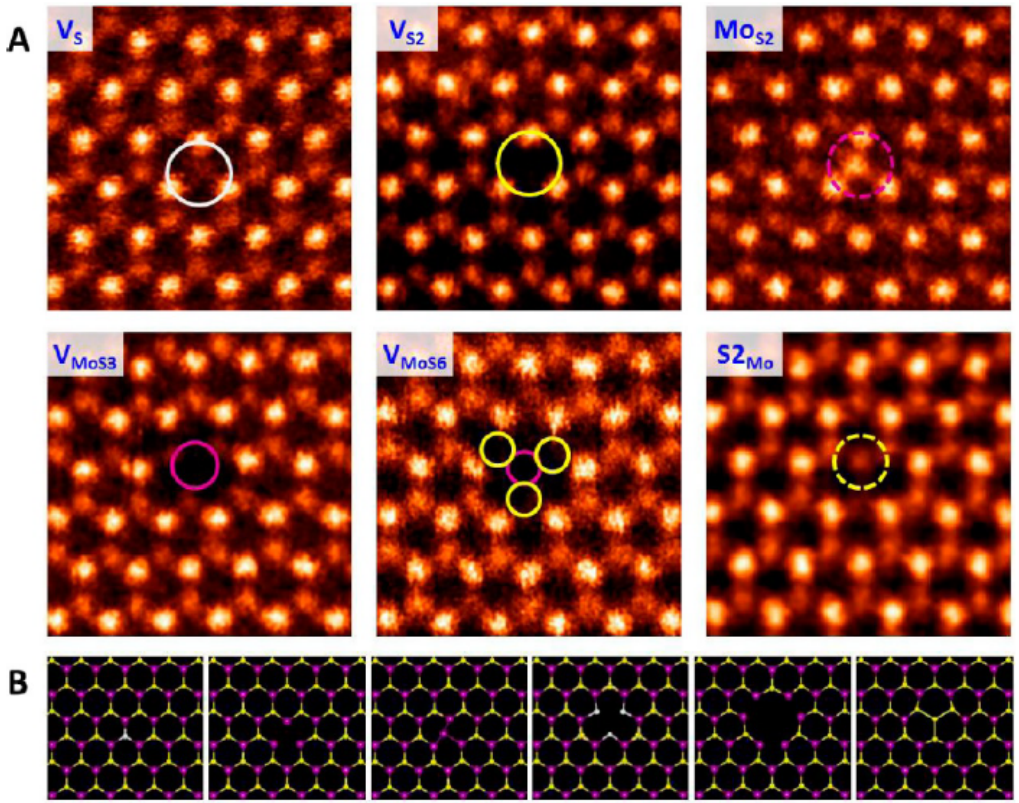




Disorder in monolayer TMDCs

- M or X centered defects/impurities
- On-site disorder with C_3 symmetry, only M-centered disorder contributes to inter-valley flipping $\langle K'|V_i^M(\mathbf{r})|K\rangle \neq 0$

  Kristen Kaasbjerg *et al.* arXiv:1708.08961 (2017)



Wu Zhou *et al.* Nano Lett. (2013)

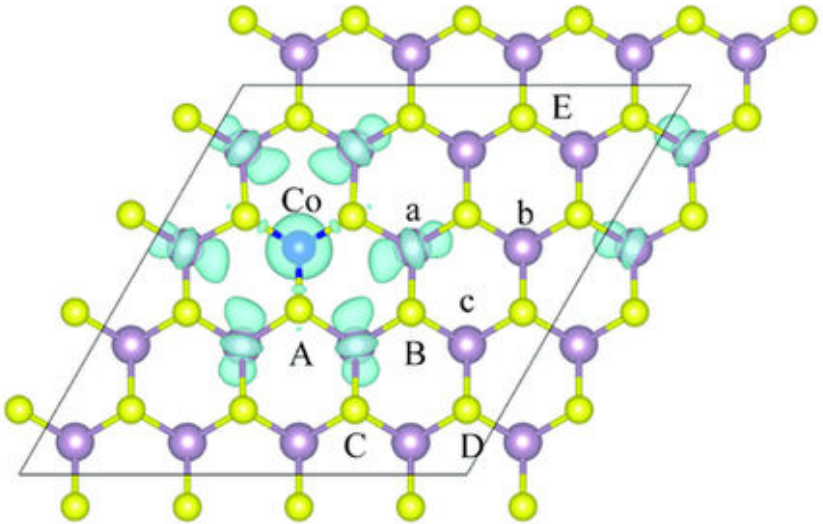
Magnetic impurity and spin-valley Zeeman interaction

- How to flip spin as well? One of the approach : Introduce the random M-centered magnetic impurities as disorder mechanism

$$H_{\text{dis}} = \sum_i J_i \delta(\mathbf{r} - \mathbf{R}_i) \otimes (\mathbf{s} \cdot \mathbf{S}) \ , \ \langle J_i \rangle = 0 \ , \ \langle J_i J_j \rangle = \langle J_i^2 \rangle \delta_{ij}$$

- Spin-valley Zeeman interaction term

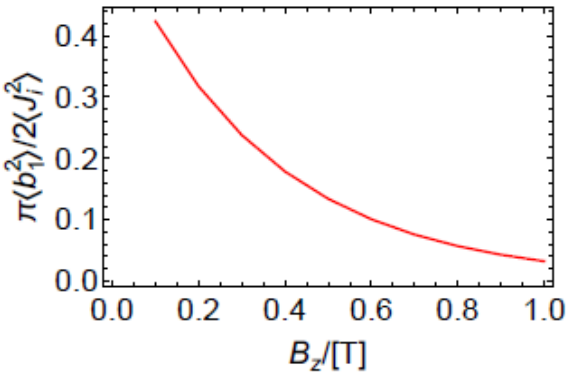
$$H_Z^{s\text{-}v} = be^{i\delta}(\tau_- s_+) \otimes S^- + \text{h.c.} = be^{i\delta}|\mathrm{K}', \uparrow\rangle\langle \mathrm{K}, \downarrow| \otimes S^- + \text{h.c.}$$

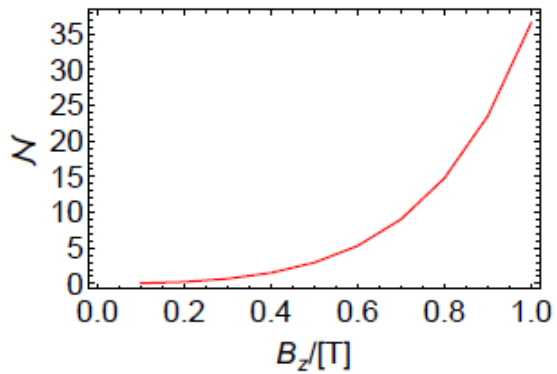


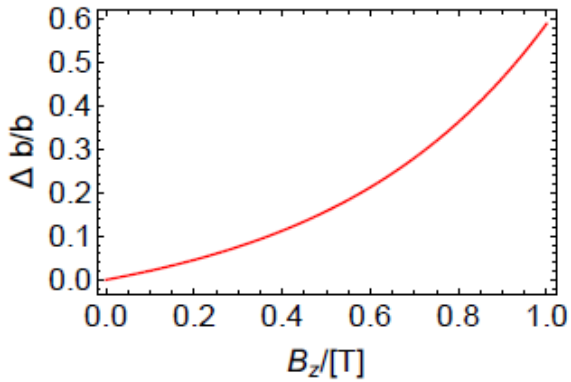
Yiren Wang *et al.* srep. (2016)

Rabi-Oscillation model

Applying AC electric field to QD $U(\mathbf{r}) \to U(\mathbf{r},t) = U(\mathbf{r} - X\sin\omega t\,\mathbf{e}_x)$, with respect to the two lowest-level states $|0,0;\mathrm{K},\downarrow\rangle$, $|0,0;\mathrm{K}',\uparrow\rangle$ under low magnetic field and in the co-moving frame $be^{i\delta} \approx b_1 e^{i\delta_1} + b_2 e^{i\delta_2}\sin\omega t$, $\mathcal{N} = \frac{\langle b_2^2\rangle}{\langle b_1^2\rangle}/(\frac{X}{l_B})^2$, denoting $b = \sqrt{\langle b_1^2\rangle}$ and $\Delta b = \sqrt{\langle b_2^2\rangle}$.



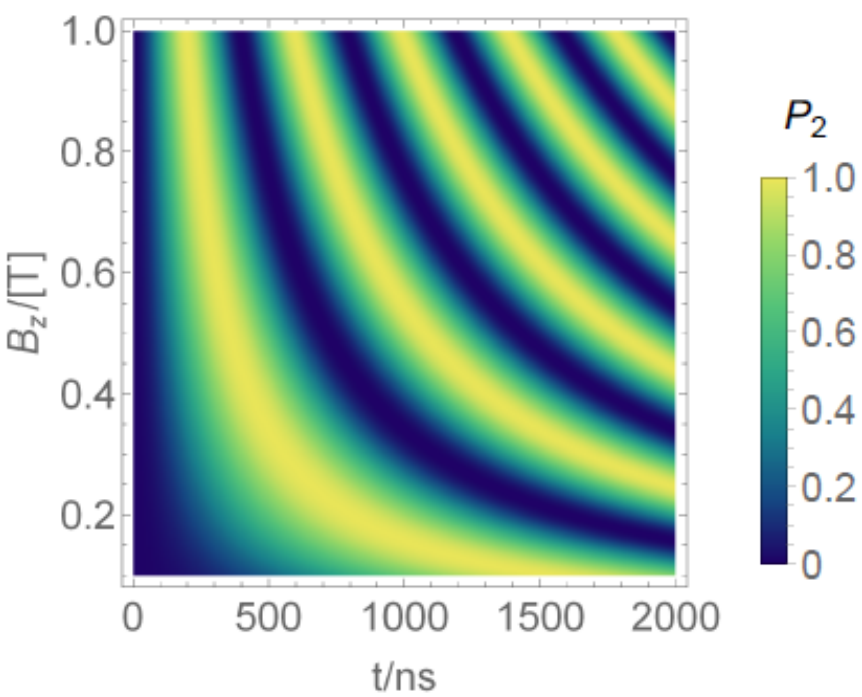




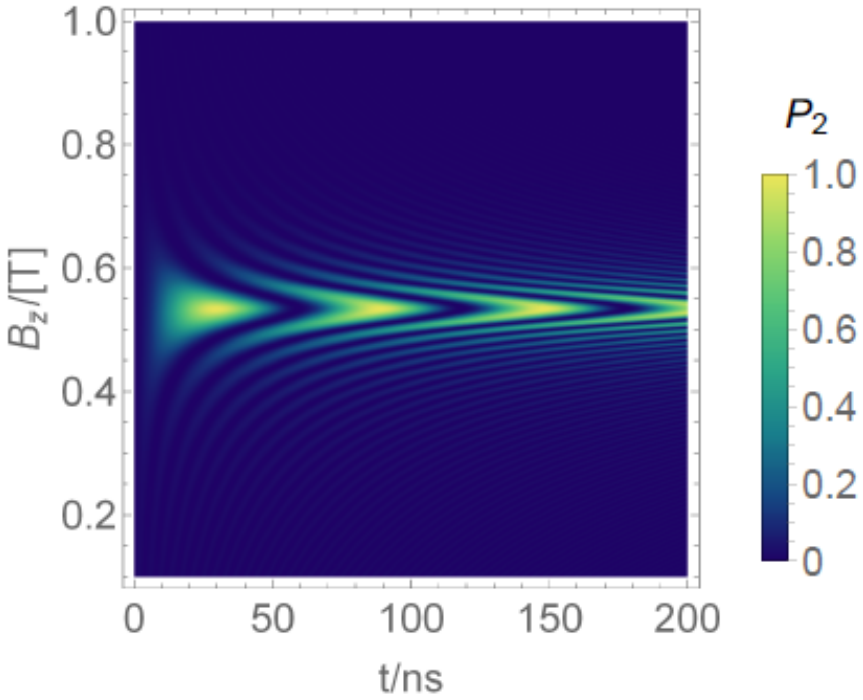
Project out the impurity spin, the Rabi-Oscillation model takes the form

$$H_{\mathrm{Rabi}} = -\frac{\hbar\omega_0}{2}\sigma^z + (b - \Delta b\sin\omega t)\sigma^x.$$

Spin-valley resonance



(a) $\Delta\omega = 0$



(b) $\omega = 2\,\mathrm{GHz}$

Summary

- monolayer TMDCs has direct band gap (good semiconductor), strong SOC, spin-valley interplay physics
- mesoscopic structure i.e. QD in TMDCs can be constructed and level structure can be tuned with magnetic field
- Spin-valley resonance induced by magnetic impurity disorder mechanism, valley qubit manipulation via quantum gate operation

The End